# PG No. 11 $C_{4h}$ $4/m$ [ tetragonal ]

* character table

| $C_{4h}$ | $1(1)$ | $2_{001}(1)$ | $4^+_{001}(2)$ | $-1(1)$ | $m_{001}(1)$ | $-4^+_{001}(2)$ |
|---|---|---|---|---|---|---|
| $A_g$ | 1 | 1 | 1 | 1 | 1 | 1 |
| $B_g$ | 1 | 1 | −1 | 1 | 1 | −1 |
| $E_g$ | 2 | −2 | 0 | 2 | −2 | 0 |
| $A_u$ | 1 | 1 | 1 | −1 | −1 | −1 |
| $B_u$ | 1 | 1 | −1 | −1 | −1 | 1 |
| $E_u$ | 2 | −2 | 0 | −2 | 2 | 0 |

* polar ↔ axial conversion

$A_g$ ($A_u$)　$B_g$ ($B_u$)　$E_g$ ($E_u$)　$A_u$ ($A_g$)　$B_u$ ($B_g$)　$E_u$ ($E_g$)

* symmetric product

| | $A_g$ | $B_g$ | $E_g$ | $A_u$ | $B_u$ | $E_u$ |
|---|---|---|---|---|---|---|
| $A_g$ | $A_g$ | $B_g$ | $E_g$ | $A_u$ | $B_u$ | $E_u$ |
| $B_g$ | | $A_g$ | $E_g$ | $B_u$ | $A_u$ | $E_u$ |
| $E_g$ | | | $A_g + 2B_g$ | $E_u$ | $E_u$ | $2A_u + 2B_u$ |
| $A_u$ | | | | $A_g$ | $B_g$ | $E_g$ |
| $B_u$ | | | | | $A_g$ | $E_g$ |
| $E_u$ | | | | | | $A_g + 2B_g$ |

* anti-symmetric product

| $A_g$ | $B_g$ | $E_g$ | $A_u$ | $B_u$ | $E_u$ |
|---|---|---|---|---|---|
| − | − | $A_g$ | − | − | $A_g$ |